

PHASE 6 – PROJECT TESTING

Project Title

Streamlining IT Procurement: Automating Standard Laptop Orders

1. Introduction

The Project Testing Phase is used to verify whether the developed IT procurement system works according to the specified requirements. The system is tested to identify errors and ensure that the main functions work correctly.

2. Testing Objectives

The main objectives are:

- Verify all major system functions.
- Identify and fix errors.
- Check input validation.
- Verify the approval workflow.
- Check database operations.
- Ensure correct request status updates.
- Confirm that the system is easy to use.

3. Testing Methods

The following testing methods are used:

- Functional Testing: Checks whether each feature works correctly.

- Input Validation Testing: Checks required and invalid inputs.
- Integration Testing: Checks communication between modules.
- Database Testing: Checks storing and retrieving data.
- User Interface Testing: Checks buttons, forms, pages, and navigation.

4. Test Cases

Test Case| Expected Result

User Login| User should login with valid details

Invalid Login| Error message should be displayed

Laptop Selection| Selected laptop details should be recorded

Submit Request| Request should be successfully submitted

Empty Required Field| Validation message should appear

Manager Approval| Request status should be updated

Request Rejection| Request should be marked as Rejected

Procurement Processing| Approved request should move to procurement

Status Tracking| Current request status should be displayed

Database Operation| Data should be stored and retrieved correctly

Note: Actual Pass/Fail status should be updated after executing each test case.

5. End-to-End Testing

The complete workflow is tested from beginning to end:

Employee Login → Laptop Request → Manager Approval → IT Review → Procurement → Order → Completed

This ensures that all modules work together correctly.

6. Error Handling

When an error is identified:

1. Identify the problem.
2. Find the cause of the error.
3. Correct the code or configuration.
4. Test the affected function again.
5. Perform regression testing to ensure that the fix does not affect other modules.

7. Final Testing Checklist

- ☐ Login tested.
- ☐ Laptop selection tested.
- ☐ Request form tested.
- ☐ Approval process tested.
- ☐ Procurement process tested.
- ☐ Database operations tested.
- ☐ Status tracking tested.
- ☐ Validation tested.
- ☐ Navigation tested.
- ☐ Major errors fixed.

8. Expected Outcome

After testing, the system should:

- Process laptop requests correctly.
- Maintain accurate request information.
- Follow the defined approval workflow.
- Update request statuses correctly.
- Store and retrieve data properly.
- Provide a reliable and user-friendly experience.

9. Conclusion

The Project Testing Phase ensures that the automated IT procurement system meets the identified requirements. Testing helps identify errors, improve system reliability, and prepare the application for final documentation and demonstration.